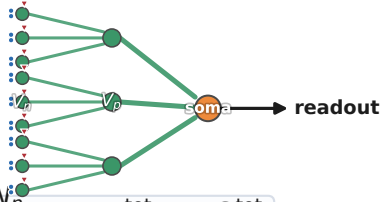


A Conductance dendritic tree

• E: positive drive

• I: conductance load



$$V_n = \frac{N_n}{G_n^{\text{tot}}}, \quad R_n^{\text{tot}} = 1/G_n^{\text{tot}}$$

inhibition raises G_n^{tot} and lowers R_n^{tot}

B Exact factorization

$$\frac{\partial L}{\partial g_i^{\text{syn}}} =$$

$$x_i R_n^{\text{tot}} (E_i - V_n)$$

local eligibility

$$\delta_n$$

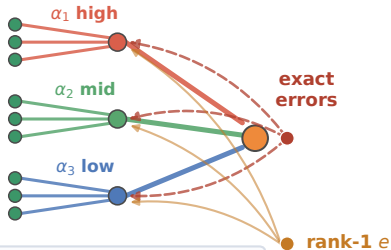
error signal

$$\text{Exact: } \delta_n = \partial L / \partial V_n$$

$$\text{LocalCA: } \delta_n \approx e_n$$

The synapse-local term is exact; the broadcast approximation is entirely in e_n .

C Path-gain field



$$\alpha_n = \prod_{\text{path}} R_n^{\text{tot}} g^{\text{den}}$$

compressible path gains make rank-1 broadcast useful

D Broadcast approximations

